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Ransomware Behaviour on Host: File System Analysis and Prevention Techniques

Abstract

Ransomware is now one of the most serious and fast developing cybersecurity threats, as it is able to encrypt data and negatively affect the critical operations of individuals, businesses and systems of countries. The study examines ransomware activity in a host environment, including the communication between ransomware and the file system and the discussion of the effective methods of preventing and mitigating it. This study addresses the following key research question: How does ransomware manipulate a host's file system during an attack, and what proactive strategies can be implemented to prevent or reduce its effect?

This research employed a multi-phase methodological framework incorporating both **behavioral examination** and **preventive assessment**. During the behavioral stage, the controlled experiment was performed under closed virtual environments to trace ransomware activity in real time. System level changes including unauthorized file encryption, privilege escalation and system directory modification were recorded by use of analytical tools like process monitors, registry trackers, and file integrity verification tools. During the preventative stage, the different defense measures, including heuristic scanning, anomaly detection, and

access control policies, were studied to determine their efficiency in curbing the recent types of ransomware.

The results demonstrate that ransomware attacks are characterized by the series of the same behavior that is usually based on the infiltration of the system, file enumeration, then encrypted and deleted recovery files including shadow copies. The researchers noted that ransomware changes file extensions, modifies registry keys in order to be persistent, and uses either symmetric or hybrid encryption algorithms to encrypt user data. Behavioral monitoring indicated some unique trends in accessing files and creating processes, which can be used as precursors to an attack. Relational analysis shows that behavioural based and machine-learning based detection systems are more effective than the traditional signature-based antivirus techniques, especially when dealing with polymorphic ransomware that continuously changes its code in order to go undetected.

The study finds that the most effective method of mitigating the risk of ransomware is a layered defense response. This involves the adoption of real time monitoring of file systems, the application of least privilege principles, regular offline file system backups, and incorporation of smart file system anomaly detection models. Understanding file system behavior of ransomware will be able to give cybersecurity professionals a basis on which they can devise proactive defense measures that can detect and prevent ransomware before it can encrypt the files.

The study is relevant to the evolving body of knowledge on cybersecurity defense in that it provides empirical evidence about the technical behavior of ransomware and demonstrates viable prevention methods that can be utilized in various computing environments.

Keywords: Ransomware, Behavioral examination, Preventive assessment, File system, Encryption, Anomaly detection, Algorithms